in_class

September 7, 2026

[20]:
```
from sklearn.cluster import KMeans
import numpy as np
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from matplotlib import pyplot as plt
%matplotlib inline
```

[21]:
```
df = pd.read_csv("income.csv")
df.head()
```

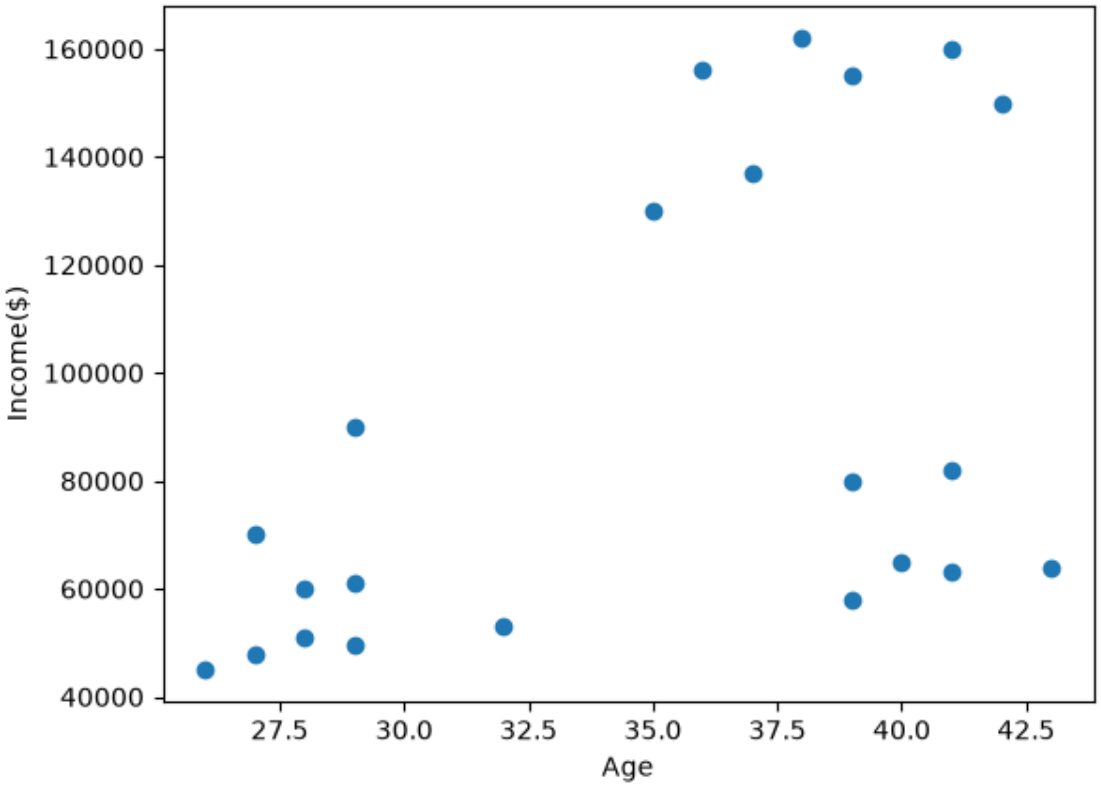
[21]:

	Name	Age	Income($)
0	Rob	27	70000
1	Michael	29	90000
2	Mohan	29	61000
3	Ismail	28	60000
4	Kory	42	150000

1 EDA

[22]:
```
plt.scatter(df.Age,df['Income($)'])
plt.xlabel('Age')
plt.ylabel('Income($)')
```

[22]:
```
Text(0, 0.5, 'Income($)')
```



2 KMean

[23]:
```
km = KMeans(n_clusters=3, init='k-means++')
y_predicted = km.fit_predict(df[['Age','Income($)']])
y_predicted
```

[23]:
```
array([1, 1, 2, 2, 0, 0, 0, 0, 0, 0, 0, 2, 2, 2, 2, 2, 2, 2, 2, 1, 1, 2],
      dtype=int32)
```

3 Clustering Output Visualization

[24]:
```
df['cluster'] = y_predicted
df.head()
```

[24]:

	Name	Age	Income($)	cluster
0	Rob	27	70000	1
1	Michael	29	90000	1
2	Mohan	29	61000	2
3	Ismail	28	60000	2
4	Kory	42	150000	0

[25]:
```
df['cluster'] = km.labels_
df.head()
```

[25]:

	Name	Age	Income($)	cluster
0	Rob	27	70000	1
1	Michael	29	90000	1
2	Mohan	29	61000	2
3	Ismail	28	60000	2
4	Kory	42	150000	0

[26]:
```
df.groupby('cluster').describe().T
```

[26]:

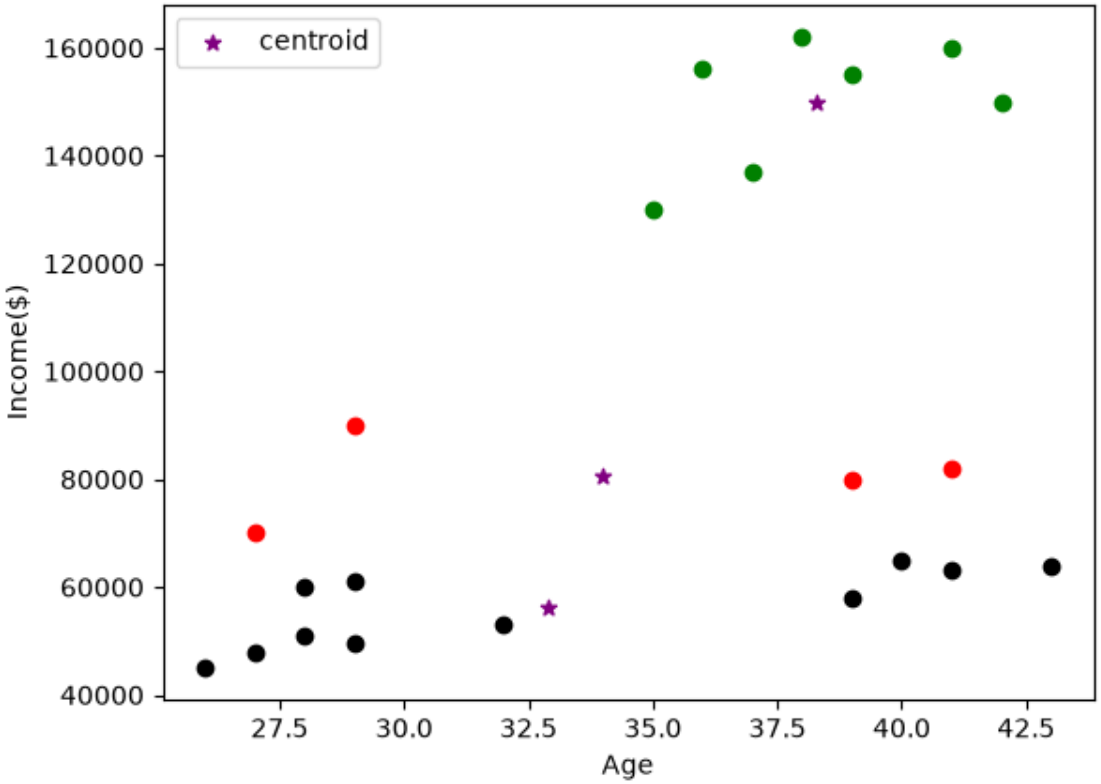
cluster		0	1	2
Age	count	7.000000	4.000000	11.000000
	mean	38.285714	34.000000	32.909091
	std	2.563480	7.023769	6.456850
	min	35.000000	27.000000	26.000000
	25%	36.500000	28.500000	28.000000
	50%	38.000000	34.000000	29.000000
	75%	40.000000	39.500000	39.500000
	max	42.000000	41.000000	43.000000
Income($)	count	7.000000	4.000000	11.000000
	mean	150000.000000	80500.000000	56136.363636
	std	12069.244660	8225.975120	7071.389217
	min	130000.000000	70000.000000	45000.000000
	25%	143500.000000	77500.000000	50250.000000
	50%	155000.000000	81000.000000	58000.000000
	75%	158000.000000	84000.000000	62000.000000
	max	162000.000000	90000.000000	65000.000000

[27]:
```
km.cluster_centers_
```

[27]:
```
array([[3.82857143e+01, 1.50000000e+05],
       [3.40000000e+01, 8.05000000e+04],
       [3.29090909e+01, 5.61363636e+04]])
```

[28]:
```
df1 = df[df.cluster==0]
df2 = df[df.cluster==1]
df3 = df[df.cluster==2]
plt.scatter(df1.Age,df1['Income($)'],color='green')
plt.scatter(df2.Age,df2['Income($)'],color='red')
plt.scatter(df3.Age,df3['Income($)'],color='black')
plt.scatter(km.cluster_centers_[:,0],km.cluster_centers_[:
 ↪,1],color='purple',marker='*',label='centroid')
plt.xlabel('Age')
plt.ylabel('Income($)')
plt.legend()
```

[28]: <matplotlib.legend.Legend at 0x7876ae08e3c0>



4 Preprocessing using min max scaler

[29]:
```
scaler = MinMaxScaler()
scaler.fit(df[['Income($)']])
df['Income($)'] = scaler.transform(df[['Income($)']])
scaler.fit(df[['Age']])
df['Age'] = scaler.transform(df[['Age']])

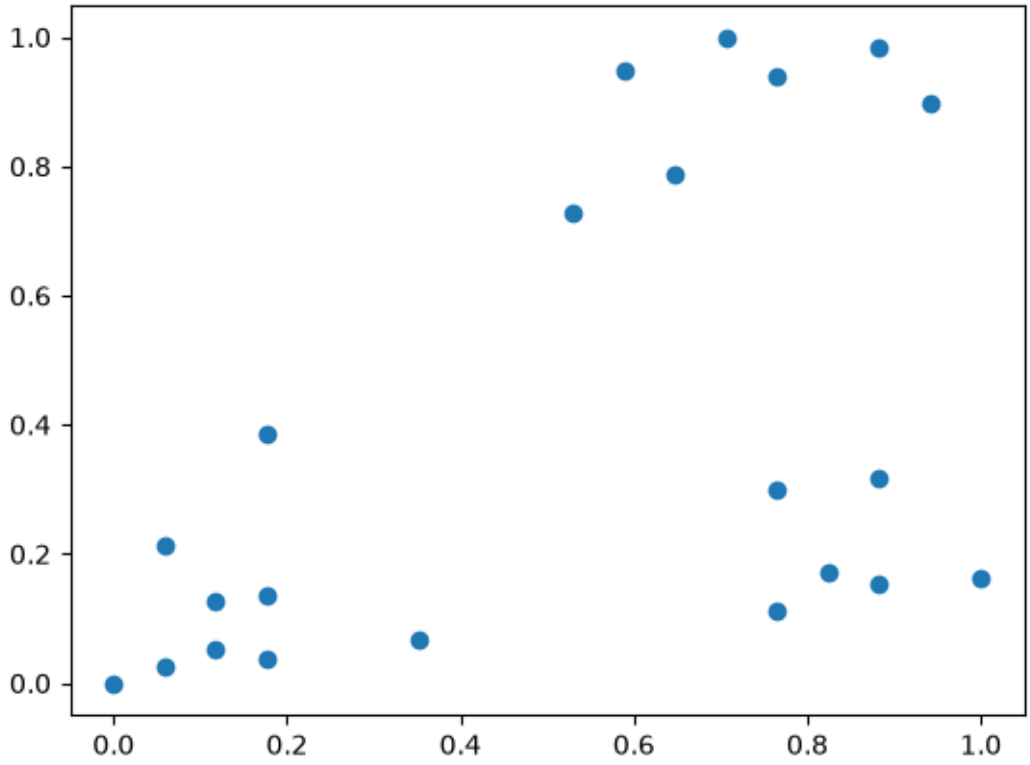
df.head()
```

[29]:

	Name	Age	Income($)	cluster
0	Rob	0.058824	0.213675	1
1	Michael	0.176471	0.384615	1
2	Mohan	0.176471	0.136752	2
3	Ismail	0.117647	0.128205	2
4	Kory	0.941176	0.897436	0

[30]:
```
plt.scatter(df.Age, df['Income($)'])
```

[30]: <matplotlib.collections.PathCollection at 0x7876ae1551d0>



[31]:
```
km = KMeans(n_clusters=3, init='k-means++')
y_predicted = km.fit_predict(df[['Age','Income($)']])
y_predicted
```

[31]:
```
array([1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2],
      dtype=int32)
```

[32]:
```
df['cluster'] = y_predicted
df.head()
```

[32]:

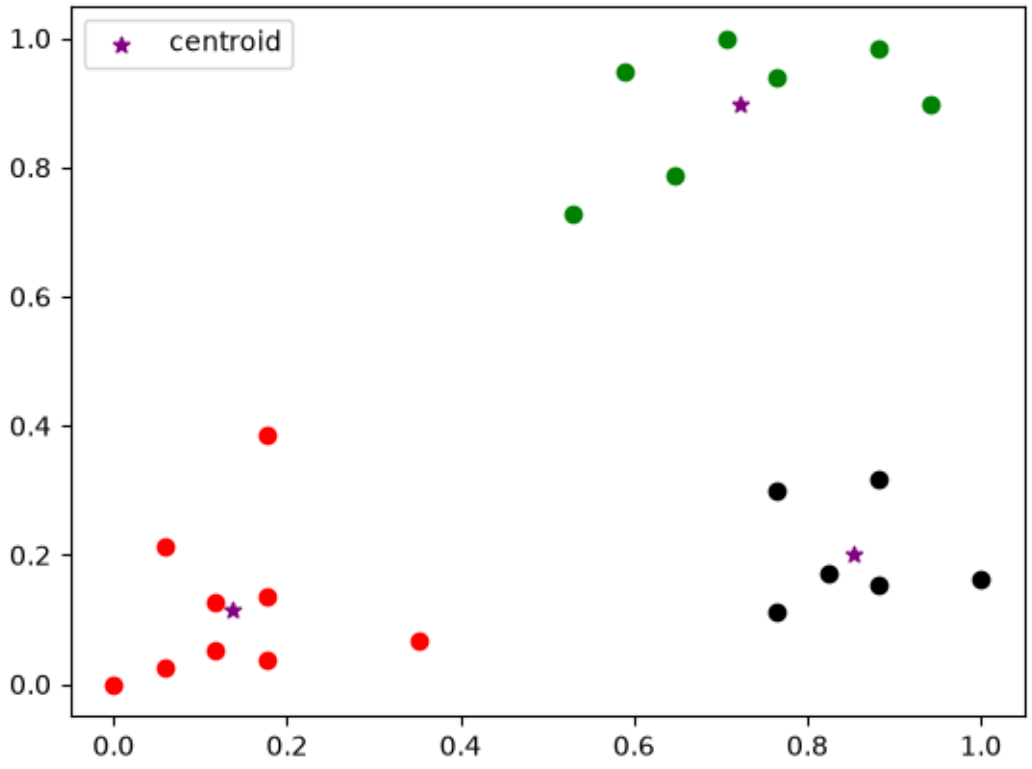
	Name	Age	Income($)	cluster
0	Rob	0.058824	0.213675	1
1	Michael	0.176471	0.384615	1
2	Mohan	0.176471	0.136752	1
3	Ismail	0.117647	0.128205	1
4	Kory	0.941176	0.897436	0

[33]:
```
km.cluster_centers_
```

[33]:
```
array([[0.72268908, 0.8974359 ],
       [0.1372549 , 0.11633428],
       [0.85294118, 0.2022792 ]])
```

[34]:
```
df1 = df[df.cluster==0]
df2 = df[df.cluster==1]
df3 = df[df.cluster==2]
plt.scatter(df1.Age,df1['Income($)'],color='green')
plt.scatter(df2.Age,df2['Income($)'],color='red')
plt.scatter(df3.Age,df3['Income($)'],color='black')
plt.scatter(km.cluster_centers_[:,0],km.cluster_centers_[:
 ↪,1],color='purple',marker='*',label='centroid')
plt.legend()
```

[34]:
```
<matplotlib.legend.Legend at 0x7876adf87cb0>
```

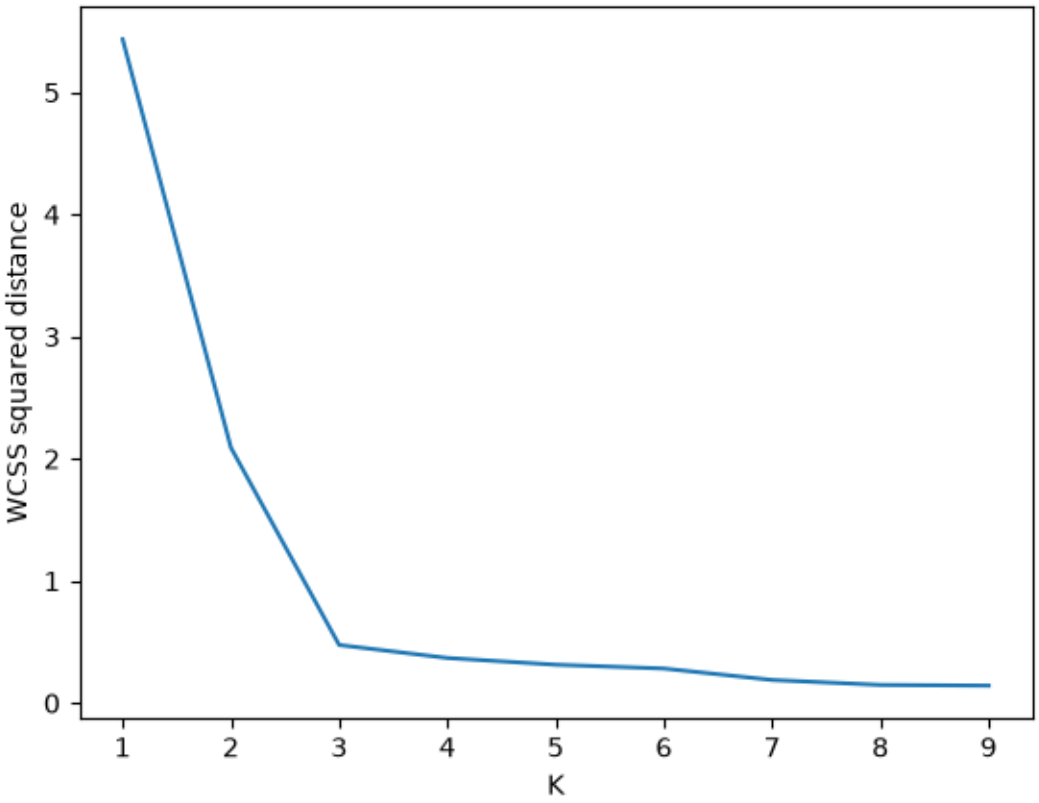


5 Elbow Plot to identify best K

[35]:
```
wcss = []
k_rng = range(1,10)
for k in k_rng:
    km = KMeans(n_clusters=k)
    km.fit(df[['Age','Income($)']])
    wcss.append(km.inertia_)

plt.xlabel('K')
plt.ylabel('WCSS squared distance')
plt.plot(k_rng,wcss)
```

[35]:
```
[<matplotlib.lines.Line2D at 0x7876ade68980>]
```



[ ]:

6 Select k

[36]:
```
for k, w in zip(k_rng, wcss):
    print(k, w)

K_SELECTED = 3
```

```
1 5.434011511988179
2 2.0911363886990784
3 0.4750783498553097
4 0.368773407644059
5 0.3138502783618938
6 0.2824643690681408
7 0.18883868819988073
8 0.1483255465451663
9 0.1425699441447823
```

7 Final KMeans model (selected k)

[37]:
```
km_final = KMeans(n_clusters=K_SELECTED, init='k-means++', random_state=42)
df['cluster'] = km_final.fit_predict(df[['Age', 'Income($)']])
df.head()
```

[37]:

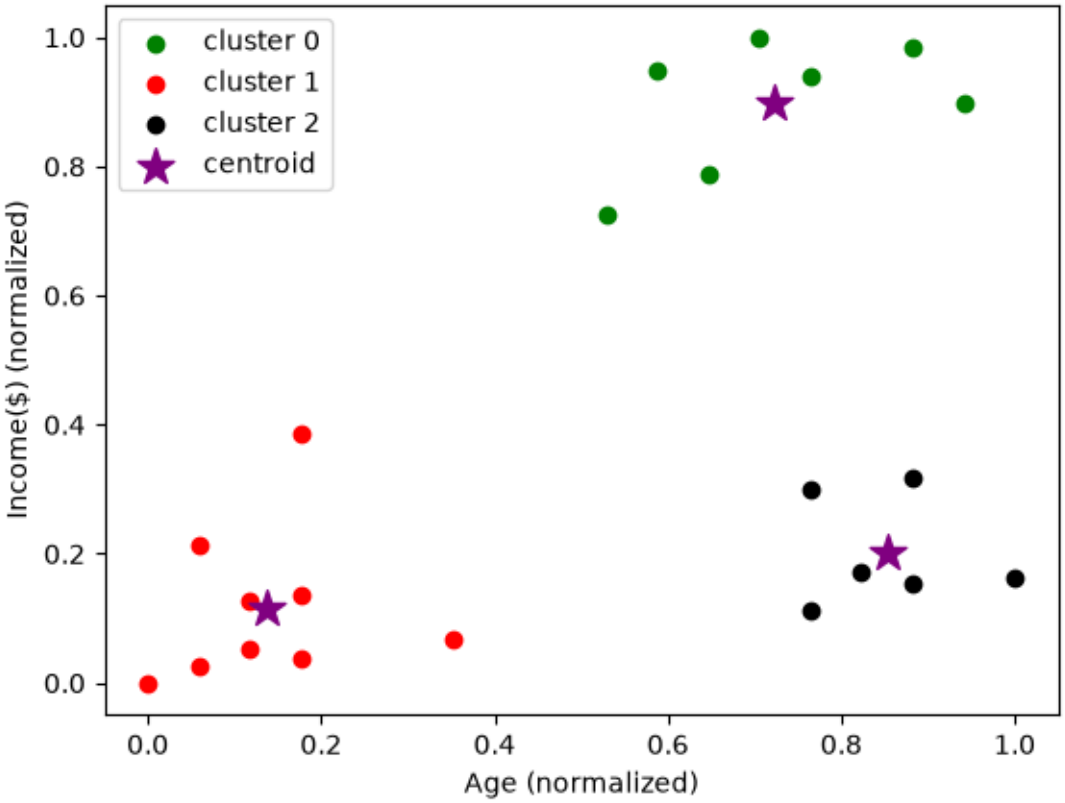
	Name	Age	Income($)	cluster
0	Rob	0.058824	0.213675	1
1	Michael	0.176471	0.384615	1
2	Mohan	0.176471	0.136752	1
3	Ismail	0.117647	0.128205	1
4	Kory	0.941176	0.897436	0

[38]:
```
km_final.cluster_centers_
```

[38]:
```
array([[0.72268908, 0.8974359 ],
       [0.1372549 , 0.11633428],
       [0.85294118, 0.2022792 ]])
```

[39]:
```
colors = ['green', 'red', 'black', 'orange', 'blue', 'brown', 'cyan',␣
 ↪'magenta', 'gray']
for c in range(K_SELECTED):
    sub = df[df.cluster == c]
    plt.scatter(sub.Age, sub['Income($)'], color=colors[c % len(colors)],␣
 ↪label=f'cluster {c}')
plt.scatter(km_final.cluster_centers_[:, 0], km_final.cluster_centers_[:, 1],
            color='purple', marker='*', s=200, label='centroid')
plt.xlabel('Age (normalized)')
plt.ylabel('Income($) (normalized)')
plt.legend()
```

[39]: <matplotlib.legend.Legend at 0x7876adebdbe0>



8 Predict new point (same training scaler)

[40]:
```
new_point_df = pd.DataFrame([{"Name": "Kevin", "Age": 35, "Income($)": 75000}])
new_point_df.to_csv("new_point.csv", index=False)
new_point_df
```

[40]:

	Name	Age	Income($)
0	Kevin	35	75000

[41]:
```
df_raw = pd.read_csv("income.csv")
scaler_age = MinMaxScaler().fit(df_raw[['Age']])
scaler_income = MinMaxScaler().fit(df_raw[['Income($)']])
```

[42]:
```
new_point = pd.read_csv("new_point.csv")
new_point_scaled = new_point.copy()
new_point_scaled['Age'] = scaler_age.transform(new_point[['Age']])
new_point_scaled['Income($)'] = scaler_income.
 ↪transform(new_point[['Income($)']])
```

```
new_point_scaled
```

[42]:

	Name	Age	Income($)
0	Kevin	0.529412	0.25641

[43]:
```
predicted_cluster = km_final.predict(new_point_scaled[['Age', 'Income($)']])[0]
print("Predicted cluster:", predicted_cluster)
```

```
Predicted cluster: 2
```

9 Interpret result

[44]:
```
print("New point:", new_point.to_dict('records')[0])
print("Predicted cluster:", predicted_cluster)
print("Cluster centroid (normalized Age, Income):", km_final.
 ↪cluster_centers_[predicted_cluster])
print()
print("Members of this cluster (original, unscaled values):")
df_raw['cluster'] = df['cluster']
print(df_raw[df_raw.cluster == predicted_cluster])
```

```
New point: {'Name': 'Kevin', 'Age': 35, 'Income($)': 75000}
Predicted cluster: 2
Cluster centroid (normalized Age, Income): [0.85294118 0.2022792 ]

Members of this cluster (original, unscaled values):
        Name  Age  Income($)  cluster
16    Dipika   40      65000        2
17  Priyanka   41      63000        2
18      Nick   43      64000        2
19      Alia   39      80000        2
20       Sid   41      82000        2
21     Abdul   39      58000        2
```